

Class 10th Social Science

Forest and Wildlife Resources

Geography





FOREST AND WILDLIFE RESOURCES



Narak! My Lord, you are the creator of music in the world of Lepchas

Oh Narak! My Lord, let me dedicate myself to you

Let me gather your music from the springs, the rivers, the mountains, the forests, the insects and the animals

Let me gather your music from the sweet breeze and offer it to you

Lepcha folk song

Flora and Fauna in India

If you look around, you will be able to find that there are some animals and plants which are unique in your area. In fact, India is one of the world's richest countries in terms of its vast array of biological diversity. This is possibly twice or thrice the number yet to be discovered. You have already studied in detail about the extent and variety of forest and wildlife resources in India. You may have realised the importance of these resources in our daily life. These diverse flora and fauna are so well integrated in our daily life that we take these for granted. But, lately, they are under great stress mainly due to insensitivity to our environment.

Activity

Find out stories prevalent in your region which are about the harmonious relationship between human beings and nature.

Conservation of Forest and Wildlife in India

Conservation in the background of rapid decline in wildlife population and forestry has become essential. But why do we need to conserve our forests and wildlife? Conservation preserves the ecological diversity and our life support systems – water, air and soil. It also preserves the genetic diversity of plants and animals for better growth of species and breeding. For example, in agriculture, we are still dependent on traditional crop varieties. Fisheries too are heavily dependent on the maintenance of aquatic biodiversity.

In the 1960s and 1970s, conservationists demanded a national wildlife protection programme. The Indian Wildlife (Protection)

Biodiversity or **Biological Diversity** is immensely rich in wildlife and cultivated species, diverse in form and function but closely integrated in a system through multiple network of interdependencies.

Chapter Summary

Forest and wildlife → why they are a resource?



- Intro
- need for Conservation.



Steps taken

- Steps taken by govt [Indian wildlife protection act, 1972, Project tiger]
- Steps taken by Community
- Types and distribution.

Steps taken by the Government for the conservation of forest and wildlife?



Fig. 2.1

Ques What are the provisions in Indian wildlife protection act 1972?

Act was implemented in 1972 with various provisions for protecting habitats. An all-India list of protected species was also published. The thrust of the programme was towards protecting the remaining population of certain endangered species by banning hunting, giving legal protection to their habitats, and restricting trade in wildlife. Subsequently, central and many state governments established national parks and wildlife sanctuaries about which you have already studied. The central government also announced several projects for protecting specific animals, which were gravely threatened, including the tiger, the one-horned rhinoceros, the Kashmir stag or hangul, three types of crocodiles – fresh water crocodile, saltwater crocodile and the Gharial, the Asiatic lion, and others. Most recently, the Indian elephant, black buck (chinkara), the great Indian bustard (godawan) and the snow leopard, etc. have been given full or partial legal protection against hunting and trade throughout India.

Project Tiger

Tiger is one of the key wildlife species in the faunal web. In 1973, the authorities realised that the tiger population had dwindled to 1,827 from an estimated 55,000 at the turn of the century. The major threats to tiger population are numerous, such as poaching for trade, shrinking habitat, depletion of prey base species, growing human population, etc. The trade of tiger skins and the use of their bones in traditional medicines, especially in the Asian countries left the tiger population on the verge of extinction. Since India and Nepal provide habitat to about two-thirds of the surviving tiger population in the world, these two nations became prime targets for poaching and illegal trading.

"Project Tiger", one of the well-publicised wildlife campaigns in the world, was launched in 1973. Tiger conservation has been viewed not only as an effort to save an endangered species, but with





Fig. 2.2: Rhino and deer in Kaziranga National Park

equal importance as a means of preserving biotypes of sizeable magnitude. Corbett National Park in Uttarakhand, Sunderbans National Park in West Bengal, Bandhavgarh National Park in Madhya Pradesh, Sariska Wildlife Sanctuary in Rajasthan, Manas Tiger Reserve in Assam and Periyar Tiger Reserve in Kerala are some of the tiger reserves of India.

- 4 The conservation projects are now focusing on biodiversity rather than on a few of its components. There is now a more intensive search for different conservation measures. Increasingly, even insects are beginning to find a place in conservation planning. In the notification under Wildlife Act of 1980 and 1986, several hundred butterflies, moths, beetles, and one dragonfly have been added to the list of protected species. In 1991, for the first time plants were also added to the list, starting with six species.
- 5
- 6

Activity

Collect more information on the wildlife sanctuaries and national parks of India and cite their locations on the map of India.

Types and Distribution of Forest and Wildlife Resources

Even if we want to conserve our vast forest and wildlife resources, it is rather difficult to manage, control and regulate them. In India,

much of its forest and wildlife resources are either owned or managed by the government through the Forest Department or other government departments. These are classified under the following categories.

- (i) **Reserved Forests:** More than half of the total forest land has been declared reserved forests. Reserved forests are regarded as the most valuable as far as the conservation of forest and wildlife resources are concerned.
- (ii) **Protected Forests:** Almost one-third of the total forest area is protected forest, as declared by the Forest Department. This forest land are protected from any further depletion.
- (iii) **Unclassed Forests:** These are other forests and wastelands belonging to both government and private individuals and communities.

Reserved and protected forests are also referred to as permanent forest estates maintained for the purpose of producing timber and other forest produce, and for protective reasons. Madhya Pradesh has the largest area under permanent forests, constituting 75 per cent of its total forest area. Jammu and Kashmir, Andhra Pradesh, Uttarakhand, Kerala, Tamil Nadu, West Bengal, and Maharashtra have large percentages of reserved forests of its total forest area whereas Bihar, Haryana, Punjab, Himachal Pradesh, Odisha and Rajasthan have a bulk of it under protected forests. All North-

50% 100%
33%
A1

100%
75%



Which of the following characteristics of the Indian Wildlife (Protection) Act are correct ?
Choose the most appropriate option.

Characteristics:

- I. To make provisions for habitat for wild animals.
- II. To publish list of protected species
- III. To ban hunting to save endangered species.
- IV. To include important subjects like forests and wildlife in the Union List.

Options:

- ☒ (A) Only I, II and III are correct,
- (B) Only I, II and IV are correct.
- (C) Only II, III and IV are correct.
- (D) Only I, III and IV are correct.

Practice Question for this Topic

CBSE Board Paper 2026 - Set [32/4/2]

Match Column - I with Column - II and choose the correct option: |

Column - I (National Park)

Column - II (States)

a. Corbett

(i) Rajasthan

b. Sundarban

(ii) Uttarakhand

c. Sariska

(iii) Kerala

d. Periyar

(iv) West Bengal

Options:

~~(A) a-ii, b-iv, c-i, d-iii~~

(B) a-ii, b-i, c-iv, d-iii

(C) a-iv, b-ii, c-iii, d-i

(D) a-ii, b-iv, c-i, d-iii

Practice Question for this Topic

CBSE Board Paper 2025 - Set [32/1/2]

Arrange the following categories of forests in India from the largest to the smallest in terms of area and choose the correct option:

I. Reserved 50%

II. Protected 33%

III. Unclassed Bacha Kundera.

Options:

(A) III, II, I

~~(B) II, III, I~~

~~(C) I, II, III~~

(D) III, I, II

Practice Question for this Topic

CBSE Board Paper 2026 - Set [32/5/2]

In which of the following States of India, has the largest area under permanent forests?

- (A) Uttar Pradesh
- (B) Madhya Pradesh
- (C) Gujarat
- (D) Maharashtra

↓
Reserved + Protected!

Practice Question for this Topic

CBSE Board Paper 2024 - Set [32/1/3]

Suggest any two measures for the conservation of forest.

2

↓
open ended Question

- illegal cutting of forest should be banned.
- Community should play important role.

Practice Question for this Topic

1.1.63

CBSE Board Paper 2024 - Set [32/1/3]

Suggest any two measures for the conservation of wildlife.

2

Open ended.

Practice Question for this Topic

CBSE Board Paper 2026 - Set [32/2/2]

Describe the main provisions of the Indian Wildlife (Protection) Act, 1972.

5

↓
⑤ Point discussed today.

Practice Question for this Topic

CBSE Board Paper 2026 - Set [32/2/2]

Describe any two characteristics of 'reserved forests' in India.

2

- Most valuable forest
- 50% of the total forest area is reserved forest.

Gharial on the brink

The gharial population has been at its lowest since the 1970s. What went wrong and what can we do?

ROMULUS WHITAKER
and JANAKI LENIN

WISPY tendrils of mist rise delicately from the water surface, tinged gold by the dawn. Your breath hangs as little clouds of vapour as you gaze upon the Girwa River on a cold winter morning. A trio of hollow clapping sounds from the other side of the river, half a kilometre away tells you that an adult male gharial is advertising his presence. It is the height of the breeding season. The place seems trapped in a time in early history when man was still clad in animal skins. It is only as the sun rises higher and burns the mist off the water that the world comes into focus with appalling clarity. The five-km stretch of the Girwa River in Katarniaghat Wildlife Sanctuary is one of the only three wild breeding sites left in the world for the most unique of all the



CRITICALLY ENDANGERED: Captive gharial at the Madras Zoo

hatched by FAO consultant Bob Buxant. When they reached a metre in length, they were released in the wild.

ability to support larger numbers of the animal. During the dry summer months, the



Bird deaths blamed on dirty Yamuna

Delhi Govt Report Points To Toxic Elements in Stagnant Water

By Nishi Sharma/TNN

New Delhi: It is official now. The recent bird deaths reported in Okhla sanctuary were because of pollution of Yamuna water and consumption of fish and not because of bird flu. The wildlife departments of UP and Delhi have sent reports to respective governments saying that since such deaths cannot be ruled out till the polluted water in this sanctuary is cleaned.

Fifty-three migratory birds were found dead in Okhla bird sanctuary earlier this month. Later UP govt. officials had accused local fishermen and claimed that they had added poison to water to kill fish. These govt. claims have been refuted.



Can you find out the reasons for the above mentioned problems?

eastern states and parts of Gujarat have a very high percentage of their forests as unclassed forests managed by local communities.

Community and Conservation

Conservation strategies are not new in our country. We often ignore that in India, forests are also home to some of the traditional communities. In some areas of India, local communities are struggling to conserve these habitats along with government officials, recognising that only this will secure their own long-term livelihood. In Sariska Tiger Reserve, Rajasthan, villagers have fought against mining by citing the Wildlife Protection Act. In many areas, villagers themselves are protecting habitats and explicitly rejecting government involvement. The inhabitants of five villages in the Alwar district of Rajasthan have declared 1,200 hectares of forest as the Bhairondev Dakav 'Sonchuri', declaring their

own set of rules and regulations which do not allow hunting, and are protecting the wildlife against any outside encroachments.

The famous **Chipko** movement in the Himalayas has not only successfully resisted deforestation in several areas but has also shown that community afforestation with indigenous species can be enormously successful. Attempts to revive the traditional conservation methods or developing new methods of ecological farming are now widespread. Farmers and citizen's groups like the **Beej Bachao Andolan** in Tehri and **Navdanya** have shown that adequate levels of diversified crop production without the use of synthetic chemicals are possible and economically viable.

In India joint forest management (JFM) programme furnishes a good example for involving local communities in the management and restoration of degraded



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CONTEMPORARY INDIA - II

Joint → Forest → Management
(They are managing jointly)
Govt + Common People

Sacred groves - a wealth of diverse and rare species

Nature worship is an age old tribal belief based on the premise that all creations of nature have to be protected. Such beliefs have preserved several virgin forests in pristine form called Sacred Groves (the forests of God and Goddesses). These patches of forest or parts of large forests have been left untouched by the local people and any interference with them is banned.

Certain societies revere a particular tree which they have preserved from time immemorial. The Mundas and the Santhal of Chota Nagpur region worship mahua (*Bassia latifolia*) and kadamba (*Anthocaphalus cadamba*) trees, and the tribals of Odisha and Bihar worship the tamarind (*Tamarindus indica*) and mango (*Mangifera indica*) trees during weddings. To many of us, peepal and banyan trees are considered sacred.

Indian society comprises several cultures, each with its own set of traditional methods of conserving nature and its creations. Sacred qualities are often ascribed to springs, mountain peaks, plants and animals which are closely protected. You will find troops of macaques and langurs around many temples. They are fed daily and treated as a part of temple devotees. In and around Bishnoi villages in Rajasthan, herds of blackbuck (chinkara), nilgai and peacocks can be seen as an integral part of the community and nobody harms them.

forests. The programme has been in formal existence since 1988 when the state of Odisha passed the first resolution for joint forest management. JFM depends on the formation of local (village) institutions that undertake protection activities mostly on degraded forest land managed by the forest department. In return, the members of these communities are entitled to intermediary benefits like non-timber forest produces and share in the timber harvested by 'successful protection'.

The clear lesson from the dynamics of both environmental destruction and reconstruction in India is that local communities everywhere have to be involved in some kind of natural resource management. But there is still a long way to go before local communities are at the centre-stage in decision-making. Accept only those economic or developmental activities, that are people centric, environment-friendly and economically rewarding.

Activity

Write a short essay on any practices which you may have observed and practised in your everyday lives that conserve and protect the environment around you.

"The tree is a peculiar organism of unlimited kindness and benevolence and makes no demand for its sustenance, and extends generously the products of its life activity. It affords protection to all beings, offering shade even to the axemen who destroy it".

Gautama Buddha (487 B.C.)



Practice Question for this Topic

CBSE Board Paper 2024 - Set [32/4/1]

Two statements are given below. They are Assertion (A) and Reason (R). Read both the statements and choose the correct option.

Assertion (A) : The famous Chipko Movement in the Himalayas successfully resisted deforestation in several areas.

Reason (R) Options : (A): Through this, community afforestation campaign with indigenous species was made enormously successful.

Both (A) and (R) are true, but (R) is not the correct explanation of (A).

☒ Both (A) and (R) are true and (R) is the correct explanation of (A).

(A) is true, but (R) is false.

(A) is false, but (R) is true.

Practice Question for this Topic

CBSE Board Paper 2026 - Set [32/1/2]

Identify the state related with the Chipko Movement:

- (A) Arunachal Pradesh
- (B) Madhya Pradesh
- (C) Uttarakhand
- (D) Bihar

Practice Question for this Topic

CBSE Board Paper 2026 - Set [32/4/2]

In which of the following regions did the "Beej-Bachao Andolan" of farmers begin?

- (A) Kumaon
- ☒ (B) Tehri
- (C) Almora
- (D) Chamoli

Practice Question for this Topic

CBSE Board Paper 2025 - Set [32/4/2]

14.03

Read the given source and answer the questions that follow:

Sacred Groves - a wealth of diverse and rare species Nature worship is an age old tribal belief based on the premise that all creations of nature have to be protected. Such beliefs have preserved several virgin forests in pristine form called Sacred Groves (the forests of God and Goddesses). These patches of forest or parts of large forests have been left untouched by the local people and any interference with them is banned. Certain societies revere a particular tree which they have preserved from time immemorial. The Mundas and the Santhal of Chota Nagpur region worship mahua (*Bassia latifolia*) and kadamba (*Anthocaphalus cadamba*) trees, and the tribals of Odisha and Bihar worship the tamarind(*Tamarindus indica*) and mango (*Mangifera indica*) rees during weddings. To many of us, peepal and banyan trees are considered sacred.

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- | | |
|---|---|
| 36.1 How do sacred groves show the inter-connectivity of spirituality and ecology ? | 1 |
| 36.2 How do tribal practices promote conservation of forests ? | 1 |
| 36.3 Why is conservation of wildlife important for all of us ? Explain. | 2 |

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Together We Can, *We Will*

